# Lab Task1: Process Management in Ubuntu

## Category 1: Process Monitoring and Information Commands

1. Display the processes currently running in the system and identify their details.
2. List all active processes in the system and observe the process identifiers.
3. Display detailed information about running processes and identify the parent-child relationships among them.
4. Monitor the system processes in real time and identify the processes consuming high CPU and memory resources.
5. Display the process hierarchy and analyze the relationship between parent and child processes.

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## Category 2: Process Creation and Execution Commands

6. Create a background process and verify its execution status.
7. Create multiple processes running in the background and observe their details.
8. Execute a process that continues running even after the terminal session is closed.
9. Identify the process ID of a running application.
10. Search for a specific running process and display its process information.

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## Category 3: Foreground and Background Process Management

11. Execute a process in the foreground and suspend its execution.
12. Display the list of active, suspended, and background processes.
13. Resume a suspended process and continue its execution in the background.
14. Move a background process to the foreground and observe its execution.
15. Terminate a running foreground process and verify its completion.

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## Category 4: Process Termination Commands

16. Create a process and terminate it using its process identifier.
17. Forcefully terminate a running process and observe the effect.
18. Create multiple instances of a process and terminate them together.
19. Terminate all processes belonging to a specific application or command.
20. Verify whether a terminated process has been removed from the process list.

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## Category 5: Process Priority Management Commands

21. Create a process and observe its default priority level.
22. Create a process with a modified priority value and analyze its behavior.
23. Change the priority of an existing process and verify the updated priority.
24. Create two processes with different priority levels and compare their execution behavior.
25. Monitor the priority changes of a process after modifying its priority value.

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